Electrical Engineering Systems

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electrical Engineering Systems (A05922)

Short Title: Electrical Engineering Systems
Faculty Science and Computing
Department Engineering Technology
Cluster Electrical Engineering
Author CLAIRE.FITZPATRICK
Credits 5 **Level:** Introductory

Description of Module / Aims

The aim of this module is to enable the student to analyse three phase circuits under balanced and unbalanced conditions. They will also study transformers and DC Machines and understand their operation and applications.

Programmes

	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	2 / 3 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 4 / M
	BEng (Hons) (`WD_ETRIC_B`)	2 / 3 / M
`ELEC-0029`	BEng in Electrical Engineering (`WD_EELCT_D`)	2 / 3 / M
`ELEC-0029`	Higher Certificate in Engineering in Electrical Engineering (`WD_EELCT_C`)	2 / 3 / M

Indicative Content

- Power in AC systems: Resistive and Reactive, Real and Apparent. Power Factor correction.
- Three phase systems: Star & Delta connections. Balanced & unbalanced loads. Power in 3 phase systems. Power factor.
- Transformers: Basic operation, Core losses, Equivalent circuits, phasor diagrams, Efficiency, Voltage regulation, Open and short circuit tests
- Machine Theory: Conversion process, magnetic field energy, Rotary motion
- DC machines: General arrangement and operation, EMF calculations, Armature reaction. Computation
- DC motors: Armature and Field connections Speed, Torque

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the differences between real and active power and perform power factor calculations.
2. Describe the generation of three phase EMFs and analyse three phase circuits.
3. Calculate transformer efficiency and voltage regulation.
4. Explain the construction and operation of DC machines.
5. Construct, analyse and simulate electrical circuits including transformers and DC machines.

Learning and Teaching Methods

- Lectures
- Tutorials
- Electrical Laboratory sessions
- Computer based simulations to investigate electrical circuit behaviour.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
Lab Report	20%	5
In-Class Assessment	10%	1,2,3,4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Hughes, E., I. McKenzie Smith and K. Brown. _Electrical and Electronic Technology_. 12th ed.. England: Pearson Education, 2016.

Supplementary Material

- Wildi, T. _Electrical Machines , Drives and Power systems_ . 6th ed.. London: Pearson International, 2014.

Requested Resources

- ENGINEERING LAB: Electrical